Flexible SABR

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1 Introduction

The SABR model [1] specifies the forward F_t and its volatility α_t by

$$\mathrm{d}F_t = \alpha_t F_t^{\beta}\,\mathrm{d}W_t, \quad \mathrm{d}\alpha_t = \nu\alpha_t\,\mathrm{d}Z_t, \quad \mathrm{d}\langle W, Z\rangle_t = \rho\,\mathrm{d}t. \tag{1}$$

2 Equivalent local volatility

Consider the SABR model with a general local volatility function C,

$$\mathrm{d}F_t = \alpha_t C(F_t)\,\mathrm{d}W_t, \quad \mathrm{d}\alpha_t = \nu\alpha_t\,\mathrm{d}Z_t, \quad \mathrm{d}\langle W, Z\rangle_t = \rho\,\mathrm{d}t, \tag{2}$$

and write f, α for the initial values. Define the *Lamperti transform* of C,

$$d(K) \coloneqq \int_f^K \frac{\mathrm{d}u}{C(u)}. \tag{3}$$

Let $P_n(T,K) \coloneqq \mathbb{E}\big[\alpha_T^n C(F_T)^n \delta(F_T - K)\big]$, so that P_0 is the marginal density of F_T and

$$\sigma^{\mathrm{loc}}(T,K)^2 \coloneqq \frac{P_2(T,K)}{P_0(T,K)} = C(K)^2\mathbb{E}\big[\alpha_T^2 \mid F_T = K\big]. \tag{4}$$

Proposition 2.1 (Local volatility for SABR)**.** *The local volatility σ^{loc} equivalent to (2) (in the sense that $\mathrm{d}\tilde{F}_t = \sigma^{\mathrm{loc}}\big(t,\tilde{F}_t\big)\,\mathrm{d}W_t$ has the same marginals as F_t) is, to first order in t,*

$$\sigma^{\mathrm{loc}}(t,K) = \alpha C(K)\sqrt{1 - 2\rho z + z^2}\left(1 - \frac{1}{2}t\,J(K)\right), \quad z \coloneqq -\nu\frac{d(K)}{\alpha}, \tag{5}$$

where $d(K) = \int_f^K \mathrm{d}u/C(u)$ is the Lamperti transform of C and the first-order coefficient is approximated, as in Hagan [1], by freezing C' at the midpoint F_{av} of the Lamperti transform, $d(F_{\mathrm{av}}) = d(K)/2$:

$$J(K) \approx \nu\alpha C'(F_{\mathrm{av}})\frac{\ln(1 - \rho z + z^2/4)}{z}, \quad J(f) = -\rho\nu\alpha C'(f). \tag{6}$$

The zeroth-order part (the case $J = 0$) is exact for the normal model $C \equiv 1$, where $J \equiv 0$:

$$\sigma^{\mathrm{loc}}(t,K) = \alpha\sqrt{1 - 2\rho z + z^2}, \quad z \coloneqq \frac{\nu}{\alpha}(f - K). \tag{7}$$

This result is due to Balland and Tran [2]; the proof is given in Section A.

2.1 Implied normal volatility

The equivalent local volatility can be converted into an implied normal volatility using the small-time expansion of Paulot [3].

Lemma 2.2 (Local volatility diffusion)**.** *Let $\mathrm{d}F_t = \sigma\tilde{C}(F_t)\,\mathrm{d}W_t$ with initial value f, and $d(f,K) \coloneqq \int_K^f \mathrm{d}u/\tilde{C}(u)$. Then the implied normal volatility is, to first order in T,*

$$\sigma_N(K) = \sigma\,\Big|\frac{f-K}{d}\Big|\,\left(1 + \frac{1}{d^2}\ln\left(\sqrt{\tilde{C}(f)\tilde{C}(K)}\,\Big|\frac{d}{f-K}\Big|\right)\sigma^2 T + o(T)\right). \tag{8}$$

2.2 Black–Scholes and Bachelier

Lemma 2.2 also gives the implied normal volatility of the Black–Scholes model, and, by inversion, the implied Black–Scholes volatility of the Bachelier model. Write $\operatorname{sinhc}(w) \coloneqq \sinh(w)/w$.

Proposition 2.3 (Black–Scholes model)**.** *Let* $\mathrm{d}F_t = \sigma F_t \,\mathrm{d}W_t$ *with* $F_0 = f$*, let* $d \coloneqq \ln(f/K)$ *and* $\operatorname{sinhc}(w) \coloneqq \sinh(w)/w$*. Then the implied normal volatility is, to first order in* T*,*

$$\sigma_N(K) = \sigma \frac{f-K}{d}\left(1 - \sigma^2 T\,\frac{\ln \operatorname{sinhc}(d/2)}{d^2} + o(T)\right), \tag{9}$$

and $\sigma_N(f) = \sigma f(1 - \sigma^2 T/24 + o(T))$*.*

Proof. Apply Lemma 2.2 with $\tilde{C}(F) = F$, so that $d(f,K) = \int_K^f \mathrm{d}u/u = \ln(f/K) = d$ and $\tilde{C}(f)\tilde{C}(K) = fK$. Since $f = Ke^d$ we have $f - K = 2\sqrt{fK}\sinh(d/2)$, hence

$$\sqrt{fK}\,\frac{d}{f-K} = \frac{d/2}{\sinh(d/2)} = \frac{1}{\operatorname{sinhc}}(d/2), \tag{10}$$

and the logarithm in (8) equals $-\ln\operatorname{sinhc}(d/2)$. At the money $d \to 0$, $(f-K)/d \to f$ and $\ln\operatorname{sinhc}(d/2) = d^2/24 + O(d^4)$ (Taylor expansion). □

Proposition 2.4 (Bachelier model)**.** *Let* $\mathrm{d}F_t = a\,\mathrm{d}W_t$ *with* $F_0 = f$*, let* $d \coloneqq \ln(f/K)$ *and* $\operatorname{sinhc}(w) \coloneqq \sinh(w)/w$*. Then the implied Black–Scholes volatility is, to first order in* T*,*

$$\sigma_B(K) = a\frac{d}{f-K}\left(1 + a^2 T\,\frac{\ln \operatorname{sinhc}(d/2)}{(f-K)^2} + o(T)\right), \tag{11}$$

and $\sigma_B(f) = (a/f)(1 + a^2T/(24f^2) + o(T))$*.*

Proof. The Bachelier model has implied normal volatility a at every strike, so σ_B is the Black–Scholes volatility σ whose implied normal volatility (9) equals a:

$$a = \sigma\frac{f-K}{d}\left(1 - \sigma^2 T\,\frac{\ln \operatorname{sinhc}(d/2)}{d^2} + o(T)\right). \tag{12}$$

To zeroth order $\sigma = \sigma_0 \coloneqq ad/(f-K)$. Solving to first order in T (series inversion in T) gives $\sigma = \sigma_0(1 + \sigma_0^2 T \ln\operatorname{sinhc}(d/2)/d^2) = \sigma_0(1 + a^2T\ln\operatorname{sinhc}(d/2)/(f-K)^2)$. At the money $(f-K)^2 = f^2d^2(1+O(d))$ and $\ln\operatorname{sinhc}(d/2) = d^2/24 + O(d^4)$ (Taylor expansion). □

2.3 SABR with general local volatility

Applying Lemma 2.2 to the zeroth-order equivalent local volatility of Proposition 2.1, i.e. $\sigma = \alpha$ and $\tilde{C}(F) = C(F)\sqrt{1 + 2\rho\nu d(F)/\alpha + \nu^2 d(F)^2/\alpha^2}$, and adding the first-order local volatility correction, gives the following for the SABR model (2) with a general local volatility function C.

Proposition 2.5 (Implied normal and Black–Scholes volatility)**.** *Consider the SABR model (2) with initial values* f*,* α*. For strike* K *and maturity* T *let* $\sigma_N(K)$ *be the implied normal (Bachelier) volatility, i.e. the volatility* σ_N *for which the Bachelier price equals* $\mathbb{E}\left[(F_T - K)^+\right]$*, and let* $\sigma_B(K)$ *be the implied Black–Scholes volatility, i.e. the volatility for which the Black–Scholes price equals* $\mathbb{E}\left[(F_T - K)^+\right]$*. Then, to first order in* T *(the term* c_2 *being an approximation, see below),*

$$\begin{aligned}
\sigma_N(K) &= \frac{\nu(f-K)}{z}\,\frac{z}{x(z)}(1 + T(c_1 + c_2)),\\
\sigma_B(K) &= \frac{\nu k}{z}\,\frac{z}{x(z)}(1 + T(c_1 + c_2 + c_3)),\\
z &= -\nu\frac{d(K)}{\alpha},\quad d(K) = \int_f^K \frac{\mathrm{d}u}{C(u)},\quad d(F_{\mathrm{av}}) = \frac{d(K)}{2},\quad k = \ln(f/K),
\end{aligned} \tag{13}$$

$$
\begin{aligned}
x(z) &= \int_0^z \frac{\mathrm{d}z'}{\sqrt{1-2\rho z'+z'^2}} = \ln\left(\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right),\\
&= \operatorname{log1p}(t), \quad t = \frac{z}{1-\rho}\left(1+\frac{z-2\rho}{\sqrt{1-2\rho z+z^2}+1}\right),\\
c_1 &= \frac{\nu^2}{x(z)^2}\ln\left(\sqrt{C(f)C(K)}\left(1-2\rho z+z^2\right)^{1/4}\frac{\alpha x(z)}{\nu(f-K)}\right),\\
c_2 &= -\frac{1}{4}\nu\alpha C'(F_{\mathrm{av}})\frac{\ln(1-\rho z+z^2/4)}{z},\\
c_3 &= \frac{\nu^2}{x(z)^2}\ln\operatorname{sinhc}(k/2), \quad \operatorname{sinhc}(w) := \sinh(w)/w.
\end{aligned}
\tag{13}
$$

The second expression for $x(z)$ avoids round-off for small z: it uses $\sqrt{1-2\rho z+z^2}-1 = z(z-2\rho)/\left(\sqrt{1-2\rho z+z^2}+1\right)$ in place of the subtraction, followed by `log1p`*. The coefficient c_1 is the first-order term of the lemma, c_2 the approximate local volatility correction, with F_{av} the midpoint of the Lamperti transform, and c_3 the correction from converting normal to Black–Scholes volatility. By the definition of z, $\nu(f-K)/z = \alpha(K-f)/d(K)$, the harmonic mean of C over $[f,K]$ times α, so this prefactor is regular at $K=f$, where it equals $\alpha C(f)$, and $z/x(z)\to 1$ there. Similarly $\nu k/z = \alpha k/(-d(K)) = \alpha\left((K-f)/d(K)\right)/\left(\sqrt{fK}\operatorname{sinhc}(k/2)\right)$, using $f-K = \sqrt{fK}k\operatorname{sinhc}(k/2)$; this is regular at $K=f$, where it equals $\alpha C(f)/f$. For $C\equiv 1$ we have $c_2 = 0$ and σ_N reduces to*

$$
\sigma_N(K) = \alpha\frac{z}{x(z)}\left(1+\frac{\nu^2 T}{x(z)^2}\ln\left(\left(1-2\rho z+z^2\right)^{1/4}\frac{x(z)}{z}\right)\right), \quad z = \nu\frac{f-K}{\alpha}. \tag{14}
$$

Proof. With $\tilde{C}(F) = C(F)\sqrt{1+2\rho\nu d(F)/\alpha+\nu^2 d(F)^2/\alpha^2}$, substitute $y = d(u)$, $\mathrm{d}y = \mathrm{d}u/C(u)$ and then $w = -\nu y/\alpha$:

$$
\tilde{d} := \int_K^f \frac{\mathrm{d}u}{\tilde{C}(u)} = \int_{d(K)}^0 \frac{\mathrm{d}y}{\sqrt{1+2\rho\nu y/\alpha+\nu^2 y^2/\alpha^2}} = \frac{\alpha}{\nu}\int_0^z \frac{\mathrm{d}w}{\sqrt{1-2\rho w+w^2}} = \frac{\alpha}{\nu}x(z). \tag{15}
$$

Hence $\alpha(f-K)/\tilde{d} = \nu(f-K)/x(z)$ and $\alpha^2/\tilde{d}^2 = \nu^2/x(z)^2$. Moreover $\tilde{C}(f) = C(f)$ since $d(f)=0$, $\tilde{C}(K) = C(K)\sqrt{1-2\rho z+z^2}$, and $|\tilde{d}/(f-K)| = \alpha x(z)/(\nu(f-K))$ (note $x(z)$ and $f-K$ have the same sign). Substituting into (8) gives (13) without its last term.

The last term is the first-order local volatility correction of Proposition 2.1, converted to an implied volatility heuristically. With $\sigma^{\mathrm{loc}}(t,K)\approx\sigma_0(K)(1-tJ(K)/2)$, averaging over $[0,T]$ multiplies σ_N by $1-TJ/4$. As Hagan [1] evaluates strike-dependent coefficients at the geometric mean $f_{\mathrm{av}} = \sqrt{fK}$, we evaluate J at the midpoint $z_0/2$ of the path, where $F = F_{\mathrm{av}}$, with C' frozen at F_{av}: $J\approx \alpha C'(F_{\mathrm{av}})\nu\left(\ln I(z_0/2)\right)/(\nu z_0/2)$. Since $z=\nu z_0$, $I(z_0/2)^2 = 1-\rho z+z^2/4$, which gives (6) and hence $c_2 = -J/4$. At the money $c_2 = \rho\nu\alpha C'(f)/4$.

For σ_B we use that the map from implied normal to implied Black–Scholes volatility at a fixed strike and maturity does not depend on how the former was obtained, so the conversion in the proof of Proposition 2.4 applies with a replaced by $\sigma_N(K)$: $\sigma_B = \sigma_N k/(f-K)(1+\sigma_N^2 T\ln\operatorname{sinhc}(k/2)/(f-K)^2)$. Inserting $\sigma_N = \nu(f-K)/x(z)(1+T(c_1+c_2))$ and keeping terms to first order in T gives the formula for σ_B, with $\sigma_N^2/(f-K)^2 = \nu^2/x(z)^2$ to leading order, which is c_3. □

Remark 2.6. The lemma is exact to first order in T for a time-homogeneous local volatility diffusion, but the equivalent local volatility of SABR depends on t and its first-order part in Proposition 2.1 is itself approximate. For general C the $O(T)$ coefficient in (13) is therefore an approximation: its last term is the correction from Proposition 2.1, converted heuristically. For $C\equiv 1$ the last term vanishes and (13) is the formula used in the normal-SABR literature.

2.4 Power-law local volatility

Now specialise to the power law $C(K) = K^\beta$ with $\beta \in [0,1]$ and write $k := \ln(f/K)$, so that $K = fe^{-k}$. The Lamperti transform is $d(K) = \big(K^{1-\beta} - f^{1-\beta}\big)/(1-\beta)$, which in terms of k is

$$d(K) = -(fK)^{(1-\beta)/2}\, k\, \mathrm{sinhc}((1-\beta)k/2), \quad \mathrm{sinhc}(w) := \sinh(w)/w, \tag{16}$$

since $f^{1-\beta} - K^{1-\beta} = 2(fK)^{(1-\beta)/2}\sinh((1-\beta)k/2)$; for $\beta = 1$ this is $-k = \ln(K/f)$. Hence in Proposition 2.5

$$z = (\nu/\alpha)(fK)^{(1-\beta)/2}\, k\, \mathrm{sinhc}((1-\beta)k/2), \quad F_{\mathrm{av}} = \left(\frac{f^{1-\beta} + K^{1-\beta}}{2}\right)^{1/(1-\beta)}. \tag{17}$$

Let $\overline{a} := \alpha F_{\mathrm{av}}^{\beta-1}$, the lognormal volatility of the local volatility function at the Lamperti midpoint. Then $F_{\mathrm{av}} = fe^{-k/2}(1 + O(k^2))$, so $\overline{a}$ agrees up to $O(k^2)$ with Hagan's $\alpha(fK)^{(\beta-1)/2}$, and $z = (\nu/\overline{a})k(1+O(k^2))$.

Proposition 2.7 (Power-law local volatility)**.** *For $C(K) = K^\beta$, let*

$$\begin{aligned} &k = \ln(f/K), \quad \mathrm{sinhc}(w) = \sinh(w)/w, \\ &F_{\mathrm{av}} = \left(\frac{f^{1-\beta} + K^{1-\beta}}{2}\right)^{1/(1-\beta)}, \quad \overline{a} = \alpha F_{\mathrm{av}}^{\beta-1}, \end{aligned} \tag{18}$$

where F_{av} is the Lamperti midpoint (and $F_{\mathrm{av}} = \sqrt{fK}$ for $\beta = 1$). Then the coefficients c_1, c_2, c_3 of Proposition 2.5 satisfy, as $k \to 0$,

$$\begin{aligned} c_1 &= (2-3\rho^2)\frac{\nu^2}{24} - \beta(2-\beta)\frac{\overline{a}^2}{24} \\ &\quad + k\left(\rho(1-\rho^2)\frac{\nu^3}{8\overline{a}} + \beta(2-\beta)\rho\nu\frac{\overline{a}}{24}\right) \\ &\quad + k^2\left((-495\rho^4 + 600\rho^2 - 104)\frac{\nu^4}{2880\overline{a}^2} - \beta(2-\beta)(4-3\rho^2)\frac{\nu^2}{288}\right. \\ &\qquad \left. - \beta(2-\beta)(19(1-\beta)^2 - 1)\frac{\overline{a}^2}{2880}\right) + O(k^3), \\ c_2 &= \rho\beta\nu\frac{\overline{a}}{4} - \beta(1-2\rho^2)\nu^2\frac{k}{16} + k^2\beta\rho(4\rho^2 - 3)\frac{\nu^3}{48\overline{a}} + O(k^3), \\ c_3 &= \frac{\overline{a}^2}{24} - \rho\nu\overline{a}\frac{k}{24} + k^2\left((4-3\rho^2)\frac{\nu^2}{288} + (20(1-\beta)^2 - 1)\frac{\overline{a}^2}{2880}\right) + O(k^3). \end{aligned} \tag{19}$$

In particular

$$c_1 + c_2 + c_3 = (2-3\rho^2)\frac{\nu^2}{24} + \rho\beta\nu\frac{\overline{a}}{4} + (1-\beta)^2\frac{\overline{a}^2}{24} + O(k), \tag{20}$$

which is the coefficient of T in Hagan's formula [1], with $\overline{a}$ as above. Hence, to first order in T and second order in k, the implied normal and Black–Scholes volatilities are

$$\begin{aligned} \sigma_N(K) &= \alpha(fK)^{\beta/2}\,\frac{\mathrm{sinhc}(k/2)}{\mathrm{sinhc}((1-\beta)k/2)}\,\frac{z}{x(z)}(1 + T(n_0 + n_1 k + n_2 k^2)), \\ \sigma_B(K) &= \alpha(fK)^{(\beta-1)/2}\,\frac{1}{\mathrm{sinhc}((1-\beta)k/2)}\,\frac{z}{x(z)}(1 + T(b_0 + b_1 k + b_2 k^2)), \end{aligned} \tag{21}$$

$$
\begin{aligned}
z &= \Big(\frac{\nu}{\alpha}\Big)(fK)^{(1-\beta)/2}k\,\mathrm{sinhc}((1-\beta)k/2), \quad x(z) = \ln\left(\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right),\\
n_0 &= (2-3\rho^2)\frac{\nu^2}{24} + \rho\beta\nu\frac{\overline{a}}{4} - \beta(2-\beta)\frac{\overline{a}^2}{24},\\
n_1 &= \rho(1-\rho^2)\frac{\nu^3}{8\overline{a}} - \beta(1-2\rho^2)\frac{\nu^2}{16} + \beta(2-\beta)\rho\nu\frac{\overline{a}}{24},\\
b_0 &= (2-3\rho^2)\frac{\nu^2}{24} + \rho\beta\nu\frac{\overline{a}}{4} + (1-\beta)^2\frac{\overline{a}^2}{24},\\
b_1 &= \rho(1-\rho^2)\frac{\nu^3}{8\overline{a}} - \beta(1-2\rho^2)\frac{\nu^2}{16} - (1-\beta)^2\rho\nu\frac{\overline{a}}{24},\\
n_2 &= (-495\rho^4+600\rho^2-104)\frac{\nu^4}{2880\overline{a}^2} + \beta\rho(4\rho^2-3)\frac{\nu^3}{48\overline{a}}\\
&\quad -\beta(2-\beta)(4-3\rho^2)\frac{\nu^2}{288} - \beta(2-\beta)(19(1-\beta)^2-1)\frac{\overline{a}^2}{2880},\\
b_2 &= (-495\rho^4+600\rho^2-104)\frac{\nu^4}{2880\overline{a}^2} + \beta\rho(4\rho^2-3)\frac{\nu^3}{48\overline{a}} + (1-\beta)^2(4-3\rho^2)\frac{\nu^2}{288} + 19(1-\beta)^4\frac{\overline{a}^2}{2880},\\
\overline{a} &= \alpha F_{\mathrm{av}}^{\beta-1}.
\end{aligned}
\tag{21}
$$

The prefactor of σ_N is $\nu(f-K)/z$, written out using $f-K=(fK)^{1/2}k\,\mathrm{sinhc}(k/2)$; it is regular at $K=f$, where it equals αf^β. Likewise the prefactor $\nu k/z$ of σ_B is $\alpha(fK)^{(\beta-1)/2}/\mathrm{sinhc}((1-\beta)k/2)$, which equals $\alpha f^{\beta-1}$ at $K=f$.

Proof. Write $\gamma := 1-\beta$, $r := \nu/\overline{a}$ and $w := \gamma k/2$. Everything below is a Taylor expansion in k or z, combined by composition, products and inversion of power series.

Exact identities. Since $F_{\mathrm{av}}^{-\gamma} = 2/(f^\gamma + K^\gamma)$ we have exactly $\overline{a} = 2\alpha/(f^\gamma+K^\gamma)$, and therefore

$$
z = rk\,\tau, \quad \tau := \frac{\tanh w}{w} = 1-\gamma^2k^2/12 + O(k^4) \quad \text{(Taylor expansion)}. \tag{22}
$$

Write the logarithm in c_1 as $L_C + L_z$ with $L_z := \ln\big(I(z)^{1/2}x(z)/z\big)$, $I(z)=\sqrt{1-2\rho z+z^2}$, and $L_C := \ln\Big(\sqrt{C(f)C(K)}\alpha z/(\nu(f-K))\Big)$. Since $\alpha z/\nu = (f^\gamma-K^\gamma)/\gamma$, a short calculation gives exactly $L_C = \ln(\mathrm{sinhc}(\gamma k/2))/(\mathrm{sinhc}(k/2))$.

Series ingredients. With $(1-2\rho z+z^2)^{-1/2} = \sum_n P_n(\rho)z^n$ (Legendre polynomials)[1] we have $x(z) = \sum_n P_{n-1}(\rho)z^n/n$. The following are then Taylor expansions in z or w (with m^{-2} obtained by series inversion):

$$
\begin{aligned}
m(z) := \frac{x(z)}{z} &= 1 + \rho\frac{z}{2} + (3\rho^2-1)\frac{z^2}{6} + O(z^3),\\
m(z)^{-2} &= 1 - \rho z + (4-3\rho^2)\frac{z^2}{12} + O(z^3),\\
L_z &= (2-3\rho^2)\frac{z^2}{24} + \rho(5-6\rho^2)\frac{z^3}{24} + e_4 z^4 + O(z^5),\\
e_4 &= \frac{-1305\rho^4+1380\rho^2-184}{2880},\\
\frac{\ln(1-\rho z+z^2/4)}{z} &= -\rho + (1-2\rho^2)\frac{z}{4} + \rho(3-4\rho^2)\frac{z^2}{12} + O(z^3),
\end{aligned}
\tag{23}
$$

[1] $(1-2\rho z+z^2)^{-1/2} = \sum_{n\ge 0} P_n(\rho)z^n$ for $|z|<1$, $|\rho|\le 1$; see [4, §18.12].

$$\begin{aligned}\ln \operatorname{sinhc}(w) &= \frac{w^2}{6} - \frac{w^4}{180} + O(w^6),\\ L_C &= (\gamma^2-1)\frac{k^2}{24} - (\gamma^4-1)\frac{k^4}{2880} + O(k^6),\\ \ln \operatorname{sinhc}(k/2) &= \frac{k^2}{24} - \frac{k^4}{2880} + O(k^6).\end{aligned} \tag{23}$$

Assembly. Since $\nu = r\overline{a}$, $\alpha C'(F_{\mathrm{av}}) = \beta\overline{a}$ and $x(z)^2 = z^2m^2 = r^2k^2\tau^2m^2$, the three coefficients are

$$\begin{aligned}\frac{c_3}{\overline{a}^2} &= (1/24 - k^2/2880)\,\tau^{-2}m^{-2},\\ \frac{c_2}{\overline{a}^2} &= -(\beta r/4)\frac{\ln(1-\rho z + z^2/4)}{z},\\ \frac{c_1}{\overline{a}^2} &= \frac{L_C + L_z}{k^2}\,\tau^{-2}m^{-2}.\end{aligned} \tag{24}$$

Now substitute $z = rk(1+O(k^2))$ and $\tau^{-2} = 1 + \gamma^2k^2/6$, so that $\tau^{-2}m^{-2} = 1 - \rho rk + B_2k^2 + O(k^3)$ with $B_2 \coloneqq (4-3\rho^2)r^2/12 + \gamma^2/6$. This gives

$$\begin{aligned}\frac{c_3}{\overline{a}^2} &= 1/24 - \rho rk/24 + ((4-3\rho^2)r^2/288 + (20\gamma^2-1)/2880)k^2 + O(k^3),\\ \frac{c_2}{\overline{a}^2} &= \beta\rho r/4 - \beta(1-2\rho^2)r^2k/16 + \beta\rho(4\rho^2-3)r^3k^2/48 + O(k^3).\end{aligned} \tag{25}$$

For c_1 write $(L_C + L_z)/k^2 = A_0 + A_1k + A_2k^2 + O(k^3)$ with

$$A_0 = \frac{\gamma^2-1}{24} + (2-3\rho^2)\frac{r^2}{24},\quad A_1 = \rho(5-6\rho^2)\frac{r^3}{24},\quad A_2 = -\frac{\gamma^4-1}{2880} - (2-3\rho^2)\gamma^2\frac{r^2}{144} + e_4r^4, \tag{26}$$

so that $c_1/\overline{a}^2 = A_0 + (A_1 - \rho rA_0)k + (A_2 - \rho rA_1 + B_2A_0)k^2 + O(k^3)$. Multiplying by $\overline{a}^2$ and using $\gamma^2 - 1 = -\beta(2-\beta)$ gives the coefficients of c_1, c_2, c_3 in (19). Adding them, the $O(1)$ part of $c_1 + c_2 + c_3$ is (20), using $1 - \beta(2-\beta) = \gamma^2$, and the $O(k)$ coefficient is $\rho(1-\rho^2)\nu^3/(8\overline{a}) - \beta(1-2\rho^2)\nu^2/16 - \gamma^2\rho\nu\overline{a}/24$. I checked these expansions against the exact coefficients numerically: the remainder after the k^2 term decays like k^3. □

Remark 2.8. *Agreement with Hagan.* Hagan's lognormal formula [1] follows from (21) by two simplifications. First, expand the zeroth-order term: its prefactor is $\nu k/z = \alpha(fK)^{(\beta-1)/2}/\operatorname{sinhc}((1-\beta)k/2)$, and since $\operatorname{sinhc}(w) = 1 + w^2/6 + w^4/120 + \ldots$ (Taylor expansion) the denominator is $1 + (1-\beta)^2k^2/24 + (1-\beta)^4k^4/1920 + \ldots$, which is Hagan's; replacing z by Hagan's variable $z_H \coloneqq (\nu/\alpha)(fK)^{(1-\beta)/2}\ln(f/K)$, which is related to z by $z = z_H\operatorname{sinhc}((1-\beta)k/2)$ so that $z/z_H = 1 + (1-\beta)^2k^2/24 + O(k^4)$, inside x changes the result only at order k^4. Second, freeze the strike dependence of the order-T coefficient: set $\overline{a} = \alpha(fK)^{(\beta-1)/2}$, which agrees with $\alpha F_{\mathrm{av}}^{\beta-1}$ up to $O(k^2)$, and drop the $O(k)$ term b_1k. This leaves

$$\begin{aligned}\sigma_B(K) = \alpha&\frac{(fK)^{(\beta-1)/2}}{1 + (1-\beta)^2k^2/24 + (1-\beta)^4k^4/1920}\,\frac{z_H}{x(z_H)}\\ &\cdot\left(1 + T\left((1-\beta)^2\frac{\overline{a}^2}{24} + \rho\beta\nu\frac{\overline{a}}{4} + (2-3\rho^2)\frac{\nu^2}{24}\right)\right),\end{aligned} \tag{27}$$

Hagan's formula. Since $b_1 \neq 0$, his order-T coefficient is accurate only to leading order in k; (21) keeps the $O(k)$ term and the exact prefactor.

The same two simplifications applied to the normal volatility: the prefactor is $\nu(f-K)/z = \alpha(fK)^{\beta/2}\operatorname{sinhc}(k/2)/\operatorname{sinhc}((1-\beta)k/2)$, whose numerator and denominator expand as $1 + k^2/24 +$

$k^4/1920 + ...$ and $1 + (1-\beta)^2k^2/24 + (1-\beta)^4k^4/1920 + ...$; freezing the order-$T$ coefficient as above gives

$$\begin{aligned}\sigma_N(K) = \alpha(fK)^{\beta/2}\frac{1 + k^2/24 + k^4/1920}{1 + (1-\beta)^2k^2/24 + (1-\beta)^4k^4/1920}\,\frac{z_H}{x(z_H)} \\ \cdot\left(1 + T\left(-\beta(2-\beta)\frac{\overline{a}^2}{24} + \rho\beta\nu\frac{\overline{a}}{4} + (2-3\rho^2)\frac{\nu^2}{24}\right)\right),\end{aligned} \tag{28}$$

the normal-volatility analogue of Hagan's formula.

At the money. For $K = f$ we have $k = 0$, $\overline{a} = \alpha f^{\beta-1}$, $\nu k/x(z) \to \overline{a}$ and $(f-K)/x(z) \to \alpha f^\beta/\nu$, so (21) gives

$$\begin{aligned}\sigma_B(f) &= \alpha f^{\beta-1}\left(1 + T\left((1-\beta)^2\frac{\alpha^2}{24f^{2-2\beta}} + \rho\beta\nu\frac{\alpha}{4f^{1-\beta}} + (2-3\rho^2)\frac{\nu^2}{24}\right)\right), \\ \sigma_N(f) &= \alpha f^{\beta}\left(1 + T\left(-\beta(2-\beta)\frac{\alpha^2}{24f^{2-2\beta}} + \rho\beta\nu\frac{\alpha}{4f^{1-\beta}} + (2-3\rho^2)\frac{\nu^2}{24}\right)\right).\end{aligned} \tag{29}$$

The order-T coefficient of $\sigma_B(f)$ is exactly the one in Hagan's at-the-money formula [1], term by term: the $(1-\beta)^2\alpha^2$, $\rho\beta\nu\alpha$ and $(2-3\rho^2)\nu^2$ terms.

2.5 Linearised displaced CEV

Now specialise to the local volatility function

$$C(K) = (f+D)^{\beta-1}(\beta K + (1-\beta)f + D), \tag{30}$$

with displacement D such that $f + D > 0$ and $\beta \in (0, 1]$. This is the first-order Taylor expansion of the displaced CEV function $(K+D)^\beta$ around $K = f$: it is affine in K, with $C(f) = (f+D)^\beta$ and constant slope $C' = \beta(f+D)^{\beta-1}$. Write $k := \ln(f/K)$ and

$$y := \beta\frac{K-f}{f+D}, \quad \ell := -\ln(1+y), \tag{31}$$

so that $C(K) = C(f)(1+y) = C(f)e^{-\ell}$; ℓ is real when $1 + y > 0$. The Lamperti transform is $d(K) = \ln(1+y)/\big(\beta(f+D)^{\beta-1}\big) = -(f+D)^{1-\beta}\ell/\beta$, hence in Proposition 2.5

$$z = \nu\frac{(f+D)^{1-\beta}}{\alpha\beta}\,\ell, \quad F_{\text{av}} = f + \frac{K-f}{1+\sqrt{1+y}}, \tag{32}$$

where F_{av} is the Lamperti midpoint $d(F_{\text{av}}) = d(K)/2$, at which $C(F_{\text{av}}) = (f+D)^\beta\sqrt{1+y}$. Let

$$\overline{a} := \alpha(f+D)^\beta\frac{\sqrt{1+y}}{F_{\text{av}}}, \quad \overline{\theta} := \beta\frac{F_{\text{av}}}{(f+D)\sqrt{1+y}}, \tag{33}$$

the lognormal volatility and the local elasticity $F_{\text{av}}C'(F_{\text{av}})/C(F_{\text{av}})$ of C at the Lamperti midpoint; for $C(K) = K^\beta$ these are the $\overline{a}$ and β of the previous subsection. Then $\alpha C'(F_{\text{av}}) = \alpha\beta(f+D)^{\beta-1} = \overline{\theta}\,\overline{a}$ and $z = (\nu/\overline{a})k(1 + O(k^2))$.

Proposition 2.9 (Linearised displaced CEV)**.** *For C as in (30), let*

$$\begin{aligned}&k = \ln(f/K), \quad y = \beta\frac{K-f}{f+D}, \quad \ell = -\ln(1+y), \quad \operatorname{sinhc}(w) = \sinh(w)/w, \\ &F_{\text{av}} = f + \frac{K-f}{1+\sqrt{1+y}},\end{aligned} \tag{34}$$

$$\overline{a} = \alpha (f+D)^{\beta} \frac{\sqrt{1+y}}{F_{\mathrm{av}}}, \quad \overline{\theta} = \beta \frac{F_{\mathrm{av}}}{(f+D)\sqrt{1+y}}, \tag{34}$$

where F_{av} is the Lamperti midpoint. Then the coefficients c_1, c_2, c_3 of Proposition 2.5 satisfy, as $k \to 0$,

$$\begin{aligned}
c_1 &= (2-3\rho^2)\frac{\nu^2}{24} - \overline{\theta}^2 \frac{\overline{a}^2}{24} + k\left(\rho(1-\rho^2)\frac{\nu^3}{8\overline{a}} + \overline{\theta}^2 \rho\nu \frac{\overline{a}}{24} \right) \\
&\quad + k^2 \left((-495\rho^4 + 600\rho^2 - 104)\frac{\nu^4}{2880\overline{a}^2} - (4-3\rho^2)\overline{\theta}^2 \frac{\nu^2}{288} + \overline{\theta}^4 \frac{\overline{a}^2}{2880} \right) + O(k^3), \\
c_2 &= \rho\overline{\theta}\nu \frac{\overline{a}}{4} - \overline{\theta}(1-2\rho^2)\nu^2 \frac{k}{16} + k^2 \overline{\theta}\rho(4\rho^2-3)\frac{\nu^3}{48\overline{a}} + O(k^3), \\
c_3 &= \frac{\overline{a}^2}{24} - \rho\nu\overline{a}\frac{k}{24} + k^2 \left((4-3\rho^2)\frac{\nu^2}{288} + \left(10\overline{\theta}^2 - 30\overline{\theta} + 19\right)\frac{\overline{a}^2}{2880} \right) + O(k^3).
\end{aligned} \tag{35}$$

Hence, to first order in T and second order in k, the implied normal and Black–Scholes volatilities are

$$\begin{aligned}
\sigma_N(K) &= \alpha (f+D)^{\beta} \sqrt{1+y}\, \mathrm{sinhc}(\ell/2)\, \frac{z}{x(z)} \left(1 + T(n_0 + n_1 k + n_2 k^2)\right), \\
\sigma_B(K) &= \alpha (f+D)^{\beta} \sqrt{(1+y)/(fK)}\, \frac{\mathrm{sinhc}(\ell/2)}{\mathrm{sinhc}(k/2)}\, \frac{z}{x(z)} \left(1 + T(b_0 + b_1 k + b_2 k^2)\right), \\
z &= -\nu \frac{(f+D)^{1-\beta}}{\alpha\beta} \ln(1+y), \quad x(z) = \ln\left(\frac{\sqrt{1-2\rho z + z^2} + z - \rho}{1-\rho} \right), \\
n_0 &= (2-3\rho^2)\frac{\nu^2}{24} + \rho\overline{\theta}\nu\frac{\overline{a}}{4} - \overline{\theta}^2\frac{\overline{a}^2}{24}, \\
n_1 &= \rho(1-\rho^2)\frac{\nu^3}{8\overline{a}} - \overline{\theta}(1-2\rho^2)\frac{\nu^2}{16} + \overline{\theta}^2\rho\nu\frac{\overline{a}}{24}, \\
b_0 &= (2-3\rho^2)\frac{\nu^2}{24} + \rho\overline{\theta}\nu\frac{\overline{a}}{4} + \left(1-\overline{\theta}^2\right)\frac{\overline{a}^2}{24}, \\
b_1 &= \rho(1-\rho^2)\frac{\nu^3}{8\overline{a}} - \overline{\theta}(1-2\rho^2)\frac{\nu^2}{16} - \left(1-\overline{\theta}^2\right)\rho\nu\frac{\overline{a}}{24}, \\
n_2 &= (-495\rho^4 + 600\rho^2 - 104)\frac{\nu^4}{2880\overline{a}^2} + \overline{\theta}\rho(4\rho^2-3)\frac{\nu^3}{48\overline{a}} \\
&\quad -(4-3\rho^2)\overline{\theta}^2\frac{\nu^2}{288} + \overline{\theta}^4\frac{\overline{a}^2}{2880}, \\
b_2 &= (-495\rho^4 + 600\rho^2 - 104)\frac{\nu^4}{2880\overline{a}^2} + \overline{\theta}\rho(4\rho^2-3)\frac{\nu^3}{48\overline{a}} \\
&\quad +(4-3\rho^2)\left(1-\overline{\theta}^2\right)\frac{\nu^2}{288} + \left(\overline{\theta}^4 + 10\overline{\theta}^2 - 30\overline{\theta} + 19\right)\frac{\overline{a}^2}{2880}.
\end{aligned} \tag{36}$$

The prefactor of σ_N is $\nu(f-K)/z$, written out using $\beta(f+D)^{\beta-1}(f-K) = (f+D)^{\beta}\left(1-e^{-\ell}\right) = (f+D)^{\beta}\sqrt{1+y}\,\ell\,\mathrm{sinhc}(\ell/2)$; it is regular at $K=f$, where it equals $\alpha(f+D)^{\beta}$. Likewise $\nu k/z = \alpha(f+D)^{\beta}\sqrt{(1+y)/(fK)}\,\mathrm{sinhc}(\ell/2)/\mathrm{sinhc}(k/2)$, which equals $\alpha(f+D)^{\beta}/f$ at $K=f$.

Proof. The expansion follows the power-law proof, with τ replaced by a function ψ and L_C by $-\ln \mathrm{sinhc}(\ell/2)$; as there, everything is a Taylor expansion in k or z combined by composition, products and inversion of power series. Write $r := \nu/\overline{a}$ and $\theta := \beta f/(f+D)$.

Exact identities. We have exactly $1+y=1-\theta\big(1-e^{-k}\big)$, $\alpha\beta(f+D)^{\beta-1}=\alpha C'(F_{\mathrm{av}})=\overline{\theta}\overline{a}$ and $z=\nu\ell/\big(\overline{\theta}\overline{a}\big)=r\ell/\overline{\theta}$, with

$$\overline{\theta}=\theta\Big(1+\big(e^{-k}-1\big)/\Big(1+\sqrt{1+y}\Big)\Big)/\sqrt{1+y}. \tag{37}$$

Write the logarithm in c_1 as L_C+L_z with $L_z=\ln\big(I(z)^{1/2}x(z)/z\big)$, $I(z)=\sqrt{1-2\rho z+z^2}$, and $L_C=\ln\big((f+D)^{\beta}\sqrt{1+y}\alpha z/(\nu(f-K))\big)$. Since $\alpha z/\nu=(f+D)^{1-\beta}\ell/\beta$, $\beta(f+D)^{\beta-1}(f-K)=(f+D)^{\beta}\big(1-e^{-\ell}\big)$ and $(f+D)^{\beta}\sqrt{1+y}=(f+D)^{\beta}e^{-\ell/2}$, we get exactly

$$L_C=\ln\frac{\ell e^{-\ell/2}}{1-e^{-\ell}}=-\ln\operatorname{sinhc}(\ell/2), \tag{38}$$

which is even in ℓ.

Series ingredients. The relation for $\overline{\theta}$ is a power series in k equal to θ at $k=0$ whose coefficients are polynomials in θ (expand $\sqrt{1-\theta(1-e^{-k})}$ binomially), so the implicit function theorem (Appendix B) solves it for θ (series inversion). The following are then Taylor expansions in k, with ψ^{-2} obtained by series inversion:

$$\begin{aligned}
\theta&=\overline{\theta}\Big(1+\big(1-\overline{\theta}\big)k/2-\overline{\theta}\big(1-\overline{\theta}\big)k^2/8+O(k^3)\Big),\\
\ell&=-\ln\big(1-\theta\big(1-e^{-k}\big)\big)\\
&=\theta k(1+(\theta-1)k/2+(1/6-\theta/2+\theta^2/3)k^2+\ldots)\\
&=\overline{\theta}k\,\psi,\\
\psi&=1-qk^2+O(k^3),\\
q&:=\big(1-\overline{\theta}\big)\big(2-\overline{\theta}\big)/24,\\
\psi^{-2}&=1+2qk^2+O(k^3),\\
L_C&=-\overline{\theta}^2k^2\psi^2/24+\overline{\theta}^4k^4/2880+O(k^6),\\
\ln\operatorname{sinhc}(k/2)&=\frac{k^2}{24}-\frac{k^4}{2880}+O(k^6).
\end{aligned} \tag{39}$$

Inserting $\theta\big(\overline{\theta},k\big)$ into ℓ, the terms of order k cancel, which is why $\psi=1+O(k^2)$ and $z=rk\,\psi$. The series for $m(z)=x(z)/z$, m^{-2}, L_z, e_4 and $\ln(1-\rho z+z^2/4)/z$ are those of the power-law proof.

Assembly. Using $\nu=r\overline{a}$ and $x(z)^2=z^2m^2=r^2k^2\psi^2m^2$, we have

$$\begin{aligned}
\frac{c_3}{\overline{a}^2}&=(1/24-k^2/2880)\,\psi^{-2}m^{-2},\\
\frac{c_2}{\overline{a}^2}&=-\big(\overline{\theta}r/4\big)\frac{\ln(1-\rho z+z^2/4)}{z},\\
\frac{c_1}{\overline{a}^2}&=\frac{L_C+L_z}{k^2}\,\psi^{-2}m^{-2},
\end{aligned} \tag{40}$$

with $\psi^{-2}m^{-2}=1-\rho rk+B_2k^2+O(k^3)$, $B_2:=(4-3\rho^2)r^2/12+2q$. This gives

$$\begin{aligned}
\frac{c_3}{\overline{a}^2}&=1/24-\rho rk/24+\Big((4-3\rho^2)r^2/288+\big(10\overline{\theta}^2-30\overline{\theta}+19\big)/2880\Big)k^2+O(k^3),\\
\frac{c_2}{\overline{a}^2}&=\overline{\theta}\rho r/4-\overline{\theta}(1-2\rho^2)r^2k/16+\overline{\theta}\rho(4\rho^2-3)r^3k^2/48+O(k^3).
\end{aligned} \tag{41}$$

For c_1 write $(L_C+L_z)/k^2=A_0+A_1k+A_2k^2+O(k^3)$ with $A_0=-\overline{\theta}^2/24+(2-3\rho^2)r^2/24$, $A_1=\rho(5-6\rho^2)r^3/24$ and $A_2=\overline{\theta}^2q/12+\overline{\theta}^4/2880-(2-3\rho^2)r^2q/12+e_4r^4$, so that $c_1/\overline{a}^2=A_0+(A_1-\rho rA_0)k+(A_2-\rho rA_1+B_2A_0)k^2+O(k^3)$. Multiplying by $\overline{a}^2$ gives (35). I checked these expansions against the exact coefficients numerically: the remainder after the k^2 term decays like k^3. □

For $D = 0$ and $\beta = 1$ we have $C(K) = K$, $\overline{\theta} = 1$ and these coefficients agree with the power-law ones at $\beta = 1$. For general β the $\overline{a}^2$ coefficient in c_1 is $-\overline{\theta}^2/24$ rather than $-\beta(2-\beta)/24$: the difference comes from $C'' = 0$ for (30), whereas $C'' \neq 0$ for F^β.

3 Conclusion

A Proof of Proposition 2.1

Throughout $W^1 = W$ and $W^2 = Z$ denote the correlated Brownian motions driving F and α.

Step 1: reduction to a point mass. Fix a strike K and define

$$z_t := \frac{1}{\alpha_t} \int_K^{F_t} \frac{\mathrm{d}u}{C(u)}. \tag{42}$$

Then $F_t = K$ if and only if $z_t = 0$, and $\partial z/\partial F = 1/(\alpha C(F))$, so

$$\delta(F_T - K) = \alpha_T C(K) \delta(z_T). \tag{43}$$

Hence

$$\begin{aligned} P_2(T,K) &= \mathbb{E}[\alpha_T C(K) \delta(z_T)], \\ P_0(T,K) &= \frac{1}{C(K)} \mathbb{E}[\alpha_T^{-1} \delta(z_T)]. \end{aligned} \tag{44}$$

Since $\alpha_T = \alpha \exp\big(-\frac{1}{2}\nu^2 T + \nu Z_T\big)$, both α_T and α_T^{-1} are exponential martingales up to a deterministic factor:

$$\alpha_T = \alpha \frac{\mathrm{d}\mathbb{Q}^\star}{\mathrm{d}\mathbb{Q}}, \quad \alpha_T^{-1} = \frac{e^{\nu^2 T}}{\alpha} \frac{\mathrm{d}\mathbb{Q}^\vee}{\mathrm{d}\mathbb{Q}}, \tag{45}$$

with

$$\frac{\mathrm{d}\mathbb{Q}^\star}{\mathrm{d}\mathbb{Q}} \mid_t = \exp\Big(-\frac{1}{2}\nu^2 t + \nu Z_t\Big), \quad \frac{\mathrm{d}\mathbb{Q}^\vee}{\mathrm{d}\mathbb{Q}} \mid_t = \exp\Big(-\frac{1}{2}\nu^2 t - \nu Z_t\Big). \tag{46}$$

Therefore

$$P_2(T,K) = \alpha C(K) \mathbb{E}^\star[\delta(z_T)], \quad P_0(T,K) = \frac{e^{\nu^2 T}}{\alpha C(K)} \mathbb{E}^\vee[\delta(z_T)]. \tag{47}$$

Step 2: dynamics of z. Writing $\mathrm{d}F = \alpha C \,\mathrm{d}W$, $\mathrm{d}\alpha = \nu\alpha \,\mathrm{d}Z$ and applying the multidimensional Itô formula to $z = g(F)/\alpha$ with $g' = 1/C$, $g'' = -C'/C^2$,

$$\begin{aligned} \mathrm{d}z_t &= \mathrm{d}W_t - \nu z_t \,\mathrm{d}Z_t - \frac{1}{2}\alpha_t C'(F_t)\,\mathrm{d}t + \nu^2 z_t \,\mathrm{d}t - \rho\nu \,\mathrm{d}t \\ &= \mathrm{d}W_t - \nu z_t \,\mathrm{d}Z_t - \frac{1}{2}\big(\alpha_t C'(F_t) + 2\rho\nu - 2\nu^2 z_t\big)\,\mathrm{d}t. \end{aligned} \tag{48}$$

Let $I(z) := \sqrt{1 - 2\rho\nu z + \nu^2 z^2}$ and $\mathrm{d}B_t := (\mathrm{d}W_t - \nu z_t \,\mathrm{d}Z_t)/I(z_t)$. Then $\mathrm{d}\langle B\rangle_t = \mathrm{d}t$, so B is a Brownian motion, and since $I'(z) = -\nu(\rho - \nu z)/I(z)$ we have $II' = \nu^2 z - \rho\nu$, giving

$$\mathrm{d}z_t = I(z_t)(\mathrm{d}B_t + I'(z_t)\,\mathrm{d}t) - \frac{1}{2}\alpha_t C'(F_t)\,\mathrm{d}t. \tag{49}$$

Note that $\mathrm{d}\langle B, Z\rangle_t = (\rho - \nu z_t)/I(z_t)\,\mathrm{d}t = -I'(z_t)\,\mathrm{d}t/\nu$.

Step 3: the two measures. Under $\mathbb{Q}^\star$, $Z_t - \nu t$ is a Brownian motion, so $B_t^\star := B_t + \int_0^t I'(z_s)\,\mathrm{d}s$ is a $\mathbb{Q}^\star$-Brownian motion and

$$\mathrm{d}z_t = I(z_t)\,\mathrm{d}B_t^\star - \frac{1}{2}\alpha_t C'(F_t)\,\mathrm{d}t. \tag{50}$$

Under $\mathbb{Q}^\vee$, $Z_t + \nu t$ is a Brownian motion, so $B_t^\vee := B_t - \int_0^t I'(z_s)\,\mathrm{d}s$ is a $\mathbb{Q}^\vee$-Brownian motion and

$$\mathrm{d}z_t = I(z_t)(\mathrm{d}B_t^{\vee} + 2I'(z_t)\,\mathrm{d}t) - \frac{1}{2}\alpha_t C'(F_t)\,\mathrm{d}t. \tag{51}$$

The two dynamics differ by a drift $2II'$. To remove it, introduce $\mathbb{Q}^{\wedge}$ with density process

$$\mathrm{d}\frac{\rho_t}{\rho_t} = -2I'(z_t)\,\mathrm{d}B_t^{\vee}, \tag{52}$$

so that $B_t^{\wedge} := B_t^{\vee} + 2\int_0^t I'(z_s)\,\mathrm{d}s$ is a $\mathbb{Q}^{\wedge}$-Brownian motion and z has under $\mathbb{Q}^{\wedge}$ the same dynamics as under $\mathbb{Q}^{\star}$. In particular $\mathbb{E}^{\wedge}[\delta(z_T)] = \mathbb{E}^{\star}[\delta(z_T)]$, and

$$\mathbb{E}^{\vee}[\delta(z_T)] = \mathbb{E}^{\wedge}[\rho_T^{-1}\delta(z_T)] = \mathbb{E}^{\wedge}[\rho_T^{-1} \mid z_T = 0]\mathbb{E}^{\star}[\delta(z_T)]. \tag{53}$$

Step 4: the density process. We look for $\rho_t = f(t, z_t)$ with

$$f(t,z) = \frac{I(z_0)^2}{I(z)^2}g(t), \quad \frac{f_z}{f} = -2\frac{I'}{I}, \quad \frac{f_{zz}}{f} = \frac{8I'^2 - 2\nu^2}{I^2}, \tag{54}$$

where we used $II'' = \nu^2 - I'^2$. Applying Itô's formula under $\mathbb{Q}^{\vee}$,

$$\mathrm{d}\frac{f}{f} = -2I'\,\mathrm{d}B_t^{\vee} + (-4I'^2 + 4I'^2 - \nu^2 + g'(t)/g(t) + \alpha_t C'(F_t)I'/I)\,\mathrm{d}t. \tag{55}$$

The diffusion coefficient is $-2I'$ as required, and the drift vanishes provided $g'(t)/g(t) = \nu^2 - \alpha_t C'(F_t)I'(z_t)/I(z_t)$. Thus

$$\rho_t = \frac{I(z_0)^2}{I(z_t)^2}e^{\nu^2 t}\exp\left(-\int_0^t \alpha_u C'(F_u)\frac{I'(z_u)}{I(z_u)}\,\mathrm{d}u\right). \tag{56}$$

For $C' = 0$ (the normal model) $\rho_t = I(z_0)^2 I(z_t)^{-2} e^{\nu^2 t}$ exactly.

Step 5: conclusion. On $z_T = 0$ we have $I(z_T) = I(0) = 1$, so by (47), (53) and (56),

$$P_0(T,K) = \frac{\mathbb{E}^{\wedge}\left[\exp\left(\int_0^T \alpha_u C'(F_u)I'(z_u)/I(z_u)\,\mathrm{d}u\right) \mid z_T = 0\right]}{\alpha C(K) I(z_0)^2}\mathbb{E}^{\star}[\delta(z_T)], \tag{57}$$

and so

$$\sigma^{\mathrm{loc}}(T,K)^2 = \frac{P_2}{P_0} = \frac{\alpha^2 C(K)^2 I(z_0)^2}{\mathbb{E}^{\wedge}\left[\exp\left(\int_0^T \alpha_u C'(F_u)I'(z_u)/I(z_u)\,\mathrm{d}u\right) \mid z_T = 0\right]}. \tag{58}$$

To zeroth order in T the conditional expectation equals one (the integrand is frozen and the integral vanishes with T; for $C \equiv 1$ it is identically one, so the result is exact). Finally $z_0 = \frac{1}{\alpha}\int_K^f \frac{\mathrm{d}u}{C(u)} = -d(K)/\alpha$, so

$$I(z_0)^2 = 1 + 2\rho\nu\frac{d(K)}{\alpha} + \nu^2\frac{d(K)^2}{\alpha^2}, \tag{59}$$

which is the zeroth-order part of (5). For $C \equiv 1$, $d(K) = K - f$ and $z = \nu(f-K)/\alpha$ gives the normal case.

Step 6: first-order correction (approximate). Write $h_u := \alpha_u C'(F_u)I'(z_u)/I(z_u)$. Expanding the conditional expectation in Step 5, $\mathbb{E}^{\wedge}\left[\exp\left(\int_0^T h_u\,\mathrm{d}u\right) \mid z_T = 0\right] = 1 + \mathbb{E}^{\wedge}\left[\int_0^T h_u\,\mathrm{d}u \mid z_T = 0\right] + \ldots$. To first order in T we evaluate the integral along the leading-order path of the bridge $z_u = z_0(1 - u/T)$ from z_0 to 0, with $\alpha_u \approx \alpha$ and $F_u = F(z_u)$; the fluctuations of z about this path are of order $\sqrt{T}$ and the drift of order T, so they enter only at higher order. Changing variable from u to z,

$$\int_0^T h_u \,\mathrm{d}u \approx \frac{T}{z_0}\int_0^{z_0} \alpha C'(F(z))\frac{I'(z)}{I(z)}\,\mathrm{d}z = TJ(K). \tag{60}$$

Hence $\sigma^{\mathrm{loc}}(T,K)^2 \approx \alpha^2 C(K)^2 I(z_0)^2/(1+TJ(K)) \approx \alpha^2 C(K)^2 I(z_0)^2(1-TJ(K))$, which is (5), with the exact path integral $J(K)$ here. The approximation (6) replaces $C'(F(z))$ by its value at the midpoint $F_{\mathrm{av}} = F(z_0/2)$ and evaluates the resulting factor $(\ln I(z_0))/z_0$ at the midpoint $z_0/2$, in the spirit of Hagan; with $z = \nu z_0$, $I(z_0/2)^2 = 1 - \rho z + z^2/4$. For C' constant this is the only approximation.

B Solving for θ in terms of $\overline{\theta}$

In the proof of Proposition 2.9 the relation $\overline{\theta} = \Phi(\theta, k)$ with

$$\Phi(\theta,k) := \theta\big(1 + \big(e^{-k}-1\big)/(1+s)\big)/s, \quad s := \sqrt{1-\theta(1-e^{-k})}, \tag{61}$$

has to be solved for θ as a series in k, with $\overline{\theta}$ held fixed. Φ is analytic near any point $\big(\overline{\theta},0\big)$ (where $s \approx 1$), $\Phi(\theta,0) = \theta$, and hence $\partial_\theta\Phi\big(\overline{\theta},0\big) = 1 \neq 0$. By the analytic implicit function theorem there is a unique analytic function $\theta(k)$ near $k=0$ with $\theta(0) = \overline{\theta}$ and $\Phi(\theta(k),k) = \overline{\theta}$, and its Taylor coefficients follow by implicit differentiation. Write $\Phi(\theta,k) = \theta + \varphi_1(\theta)k + \varphi_2(\theta)k^2 + O(k^3)$, where the φ_n are polynomials in θ (expand s binomially); here

$$\varphi_1(\theta) = -\theta\frac{1-\theta}{2}, \quad \varphi_2(\theta) = \theta(1-\theta)\frac{2-3\theta}{8}. \tag{62}$$

Differentiating $\Phi(\theta(k),k) = \overline{\theta}$ once and twice in k,

$$\partial_\theta\Phi\,\theta' + \partial_k\Phi = 0, \quad \partial_\theta\Phi\,\theta'' + \partial_{\theta\theta}\Phi\,\theta'^2 + 2\partial_{\theta k}\Phi\,\theta' + \partial_{kk}\Phi = 0, \tag{63}$$

and at $(\theta,k) = \big(\overline{\theta},0\big)$ we have $\partial_\theta\Phi = 1$, $\partial_k\Phi = \varphi_1$, $\partial_{\theta k}\Phi = \varphi_{1'}$, $\partial_{kk}\Phi = 2\varphi_2$ and $\partial_{\theta\theta}\Phi = 0$. Hence

$$\theta'(0) = -\varphi_1\big(\overline{\theta}\big) = \overline{\theta}\frac{1-\overline{\theta}}{2}, \quad \frac{1}{2}\theta''(0) = \varphi_1\big(\overline{\theta}\big)\varphi_{1'}\big(\overline{\theta}\big) - \varphi_2\big(\overline{\theta}\big) = -\overline{\theta}^2\frac{1-\overline{\theta}}{8}, \tag{64}$$

which gives

$$\theta(k) = \overline{\theta}\big(1 + \big(1-\overline{\theta}\big)k/2 - \overline{\theta}\big(1-\overline{\theta}\big)k^2/8 + O(k^3)\big). \tag{65}$$

Each Taylor coefficient is a polynomial in $\overline{\theta}$, because $\partial_\theta\Phi = 1$ at $k=0$ (so no division occurs) and all derivatives of Φ at $\big(\overline{\theta},0\big)$ are polynomials in $\overline{\theta}$.

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